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Ivar Cashin Eriksson

Dear Members of the Selection Committee,

I am applying for the PhD scholarship in Machine Learning for Quantum Chemistry of Complex Reactions at DTU Energy. After seven years working with machine learning in research and industry, I want to return to research. I enjoy working on difficult scientific problems with people I can learn from. I wish to pursue problems that require time and dedication to investigate why a method works, where it fails, and how it can be improved.

What attracts me to this project is the opportunity to develop machine learning methods in close connection with a physical problem. The project would allow me to build on my experience in machine learning while returning to my physics background and learning more about quantum chemistry and quantum computing. Representing electronic structure consistently as a reaction progresses, and identifying which degrees of freedom matter, are questions I would like to understand and work on. In addition, I am interested in how the resulting methods and datasets could support quantum-computing studies. The collaboration with NQCP would provide an environment to learn about that connection through concrete research.

At RaySearch Laboratories, I researched methods for automated radiotherapy planning, first through my master's thesis and then as a full-time researcher. I developed anatomical representations from segmented CT scans using autoencoders and used sparse Gaussian processes to predict dose-volume histograms and spatial dose distributions. I also worked on lexicographic optimisation methods to better represent clinical priorities. I enjoyed the technical work and environment equally: discussing difficult questions with medical physicists and researchers, testing ideas, and learning enough about the underlying problem to make useful contributions. In addition, improving cancer treatment gave the work a purpose.

Since then, I have gained experience taking responsibility for larger computational systems. At Valcon, I led machine learning projects from problem formulation through to implementation and deployment. At Signum Life Science, I now work on the architecture and development of a forecasting platform supporting thousands of independently trained models. These roles have taught me to structure complex software, design evaluation procedures, investigate failures, and work independently while coordinating with others. They have also made me realise that I want to work on more challenging scientific questions that require sustained investigation in an environment where learning and careful thought are central.

My strongest experience is in machine learning, mathematical modelling, and Python development. I would need to learn the quantum-chemical methods involved, but my training in physics and mathematics, combined with my experience learning new domains, gives me confidence that I can do so. My degree in Engineering Physics, with a specialisation in Statistical Learning and Data Analytics, gives me a strong foundation to build on. I would bring experience developing and evaluating methods, building the software needed to use them, and working with specialists whose knowledge complements my own.

I welcome the opportunity to discuss the project and where my background could be most useful.

Kind regards,
Ivar Cashin Eriksson